

LUKE SAUTER, PhD

L T C O L , U S A F , Top Secret, SSBI Dated 12 Nov 2013
LSauter@alum.mit.edu

10 Primrose St Unit 833
Palmer Lake, CO, 8013

719-333-9730 (w), 719-671-9781 (c)

EDUCATION

- PhD, Electronic Engineering, University of Surrey, UK, 2012
- SM, Aero/Astro Engineering, Massachusetts Institute of Technology, MA, 2004, *MIT Draper Fellow*
- BS, Astronautical Engineering, United States Air Force Academy, CO 2002, *Distinguished Graduate*

SELECTED ACCOMPLISHMENTS

- Legislative liaison: Responsible for Congressional engagement of a \$20B+ portfolio including five intel platforms
- Program manager: \$330M advanced imagery system acquisition, leading 170+ personnel on four unique contracts
- Led 12-member team in the acquisition of a \$194M advanced electro-optical payload for tech insertion in a \$5B+ national system. Directs complex 4-contract execution along a new acquisitions paradigm for DNI's highest priority
- Led 25-member team in the development of a \$55M small satellite. Worked with teams across multiple divisions. Effort was rated as one of the top 5 AFSPC space experiments. Also managed \$11M in additional tech development
- Piloted 7-mbr AFRL team--\$300M USECAF requested space control demo, gives 15x more SSA capability in orbit
- Directed \$500K sounding rocket program as the capstone mentor. Directed the design, development, test, integration, & launch. Led 26 cadets & 3 international cadets in the launch of the Academy's first staged sounding rocket--flew SERB requested payload--Set collegiate world record for altitude, 106km--Lauded by Deputy USecAF
- Responsible for execution of 24 counter-space & nano-satellite tech efforts worth \$25M. Developed combined comm/optical device 3x lighter than state of the art. Designed and built ground station and space data fusion center
- Administered day-to-day operations for a 36-member academic department. Responsible for ensuring the effective communication between O-7, O-6 Dept Head and faculty. Selected to instruct course at French Air Force Academy

PROFESSIONAL EXPERIENCE

Jun 17 – Present	<u>Assistant Professor of Astronautics, USAF Academy, CO:</u> Assistant Professor, division course director, department personnel officer, and department accreditation lead.
Jun 16 – Jun 17	<u>Legislative Liaison & Senior Air Force Officer, BPO/OCPA, NRO, VA:</u> Developed & executed corporate strategies to support Cabinet-level policy decisions to IC, DoD, OMB and Congress. Responsible for Congressional engagement in support of the NRO's GEOINT Directorate covering a \$20B+ FYDP portfolio including five ACAT-1 intelligence platforms.
Jun 14 – Jun 16	<u>Chief, ISR Mission Payloads Branch, IMINT/ESPO, NRO, VA:</u> Led acquisition and delivery of a top priority \$330M advanced electro-optical payload for a \$10B+ ACAT-1D national system. Responsible for 170+ personnel and cost, schedule, performance, tech insertion, and security.
Oct 12 – May 14	<u>Deputy Program Manager & Systems Engineer, IMINT/R&T, NRO, VA:</u> Responsible for acquisition of a top priority \$194M advanced electro-optical payload for \$5B+ national system. Leading 12-member team syncing enterprise efforts across 15+ orgs for a new imaging capability.
Sep 09 – Sep 12	<u>PhD Student, University of Surrey, Guildford, UK:</u> Developed novel flight software for rapid collision-free formation-flight. Enabling autonomous clustered satellite ops for DARPA missions
Jun 07 – Sep 09	<u>Executive Officer & Assistant Professor of Astronautics, USAF Academy, CO:</u> Faculty lead for \$230k capstone program, FalconLAUNCH. Flew DoD SERB reviewed payload at hypersonic speeds. Responsible for instruction of 255 cadets in astronautics. Exec for 36 member department
Jan 05 – Jun 07	<u>Space Superiority Systems Program Manager, Kirtland AFB, NM:</u> Deputy of the space superiority office & program manager for the \$55M "ANGELS" highly capable small satellite demonstration mission. Responsible for execution of 24 additional counter-space tech efforts. Development team for USecAF requested concept for critical need, \$300M multi-satellite effort.
Jun 04 – Jan 05	<u>Deputy, Advanced Spacecraft Mechanisms Group, Kirtland AFB, NM:</u> Responsible for 8 personnel, national meeting coordination, & budget execution. Miniature Vibration Isolation System payload manager for TacSat-2 satellite. Lead for \$5M in 11 tech development contracts.
Aug 02 – Jun 04	<u>Draper Fellow, Massachusetts Institute of Technology, Cambridge, MA:</u> Thesis work in satellite constellation design for mid-course ballistic missile defence. Application to MDA study

TRAINING

- Air Command and Staff College, Schools list, In-Residence/Correspondence Completion, 2012
 - Space 200, Space Professional Level-2 certification, NSSI 2008
 - Squadron Officer's School In-Residence/Correspondence, Top-Third Graduate (Used in place of D.G. at the time)
 - Certified: Engineering Level-3, Program Management Level-2, Test Level-1
-

AWARDS

- Arthur S. Flemming Award 2008, Air Force winner & National Finalist, engineering excellence in Applied Science, Engineering and Mathematics
- Roger R. Bate Award for the Outstanding First Year Instructor of Astronautics 2008; John P. Wittry Award for top instructor of Engineering Design 2009; Top Instructor, Department of Astronautics, USAFA, 3x, 2007-09.